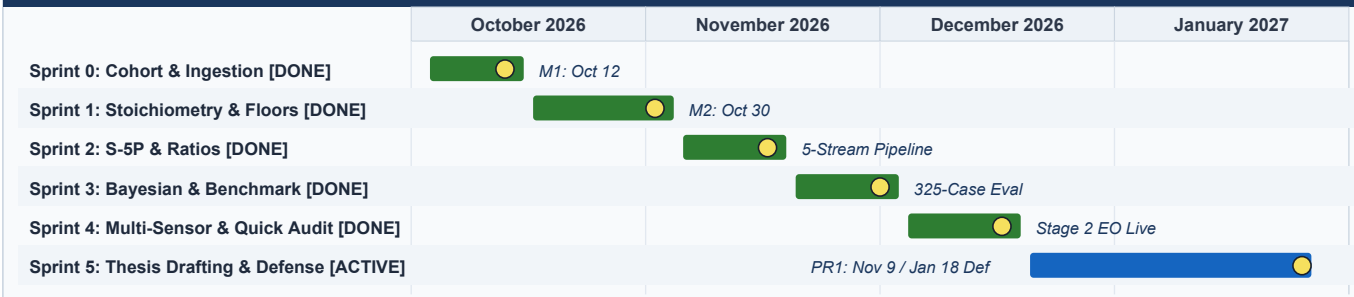
VeriCBAM: Capstone Project Work Plan, Risk Matrix & Execution Roadmap

A Data-Driven Decision-Support Tool for Screening Carbon Border (CBAM) Emissions Reports

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Degree Program:	Master of Science in Data Science (M.Sc. DSc)	**Faculty Supervisor:**	Dr. Humera Noor | Winter Semester 2026/2027

VeriCBAM Execution Gantt Chart & Milestone Timeline (Winter Semester 2026/2027)



1. Updated Execution Schedule & Milestone Status (Status: November 2026)

Sprint / Phase	Planned Dates	Key Completed Technical Tasks	Deliverable & Current Status
Sprint 0: Cohort & Ingestion	Oct 03 – Oct 14	Scoped project; ingested and cleaned 177 plant-years across 30 EU cement/steel facilities from EEA E-PRTR v16 and GEM trackers.	M1: Topic Approval (Oct 12) `[COMPLETED]`
Sprint 1: Stoichiometry & Planning	Oct 15 – Oct 26	Implemented calcination and steel reduction mass-balance floors (`cement_models.py`, `steel_models.py`); boundary gas-export rules.	M2: Proposal Approved (Oct 30) `[COMPLETED]`
Sprint 2: Satellite & Baselines	Oct 27 – Nov 04	Ingested Sentinel-5P TROPOMI NO□; built leave-one-year-out NO_x/CO□ ratios and one-sided robust pooled-MAD longitudinal drift scores.	5-Stream Data Pipeline `[COMPLETED]`
Sprint 3: Modeling & Evaluation	Oct 28 – Nov 05	Built Bayesian Log-Odds Fusion (Model E1) & GroupKFold Logistic Triage (Model E3); evaluated 325-case graded benchmark with 1,000 bootstrap CIs.	325-Case Benchmark `[COMPLETED]`
Sprint 4: Multi-Sensor & Dashboard	Nov 01 – Nov 05	Built live 8-tab Streamlit Cockpit with Tab 1 Quick Audit & PDF Dossier; integrated Stage 2 Multi-Sensor EO (Landsat 30m TIR, VIIRS 375m, Planet VHR).	Live App & Quick Audit `[COMPLETED]`
Sprint 5: Thesis & Defense	Nov 06 – Jan 18	Master thesis drafting (Chapters 1–6), code reproducibility audit, external verifier outreach, and final oral defense presentation.	PR1 (Nov 09) / Final Sub (Jan 04) `[ACTIVE]`

2. Project Risk Management & Resolution Matrix

Identified Technical Risk	Severity	Likelihood	Impact on Project	Implemented Mitigation & Current Resolution
Cloud Cover in Sentinel-5P Satellite	High	High	Trace-gas data gaps during northern European winter.	Resolved: Weighted satellite evidence by cloud fraction (C_{sat} = 1 - f_{cloud}) and integrated cloud-resilient thermal infrared (Landsat/VIIRS).
Off-Site Process Gases in Steelworks	High	Medium	False physical violations on blast furnace plants exporting gas to power plants.	Resolved: Mapped plant boundaries in GEM registry; set `stoich_applicable = False` for gas-exporting plants to eliminate false alarms.
Data Leakage in Machine Learning	Critical	Low	Artificially inflated accuracy from training and testing on the same plant.	Resolved: Enforced strict 5-Fold `GroupKFold(groups=facility_id)`, ensuring all years of any plant are strictly isolated between folds.
Confidentiality of Overseas Fraud	High	Medium	Real non-EU fraudulent CBAM reports are legally confidential.	Resolved: Built a 325-case controlled benchmark combining 177 real compliant years with 148 graded synthetic understatements (5%–50%).

3. Core Deliverables & Current Completion Status

1. VeriCBAM Python Engine (`src/`): 100% complete and passing 30/30 unit tests (`pytest`).
2. Interactive Cloud Decision Cockpit: Live at https://vericbam-uqpz4jrnpteqetfvjxtslh.streamlit.app/ featuring the new Tab 1 Quick Audit & 1-Click PDF Dossier Generator.
3. Stage 2 Multi-Sensor Satellite Dataset: Cached 30-facility dataset combining USGS Landsat-8/9 30m TIR Δ T, NASA FIRMS VIIRS 375m operating calendars, and Planet VHR freight contrast.
4. Master Thesis Manuscript: Drafting in progress following academic structure under bi-weekly supervision of Dr. Humera Noor.

4. Remaining Roadmap Towards Master Thesis Submission (Winter 2026/2027)

- Nov 09, 2026: Progress Report 1 Submission (Demonstrating completed Stage 1 & Stage 2 empirical evaluation).
- Nov 23, 2026: Progress Report 2 Submission (Draft Thesis Methodology & Empirical Results chapters).
- Dec 07 – Dec 21, 2026: Progress Reports 3 & 4 (Complete manuscript review, conclusion, and policy recommendations).
- Jan 04, 2027: Final Master Thesis Submission (Binding university submission).
- Jan 18, 2027: Master Thesis Oral Defense & Examination.